

algo gate

CSE, IT, DA

DAC

Get video of
the DAC

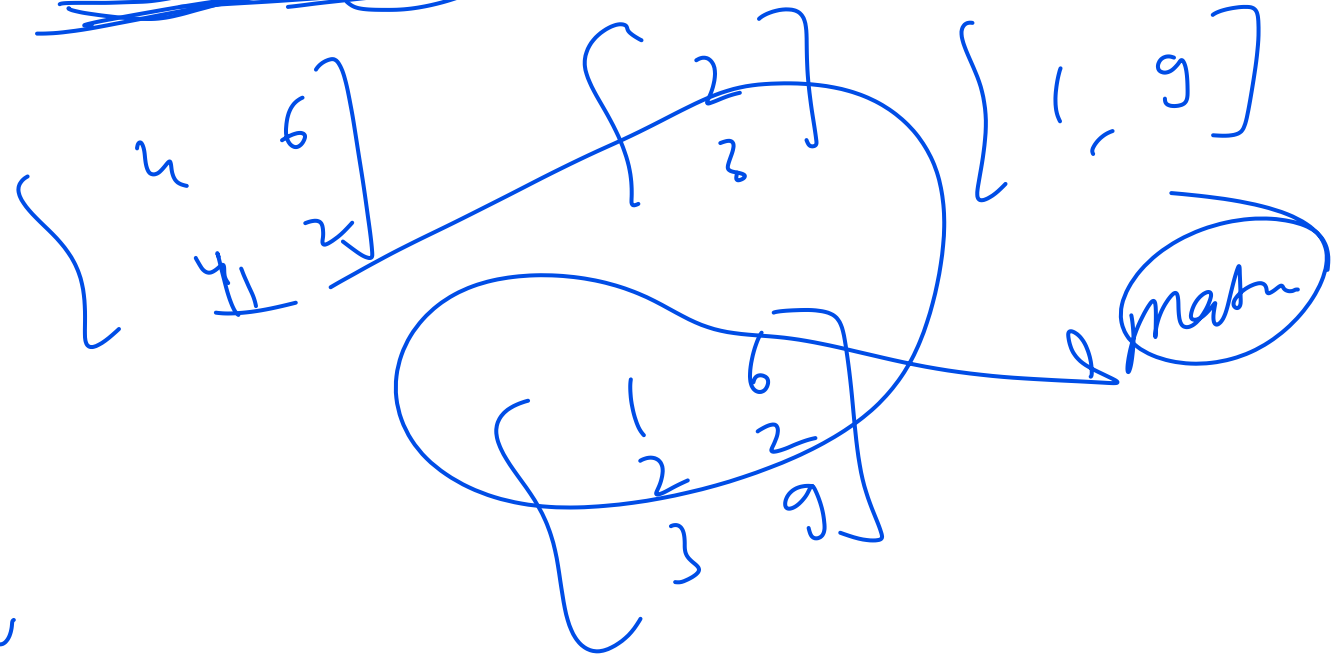
Soehing

Lec_47

Strassen's matrix multiplication

Placeme

Defermu



Matrix Addition : $O(n^2)$
Matrix multiplication : $O(n^3)$, without DAC

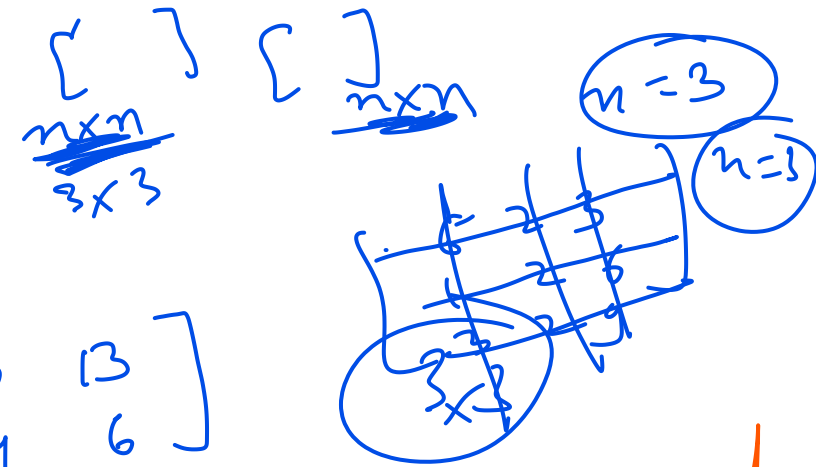
WITH DAC:

Example:

$A = \begin{bmatrix} 2 & 3 \\ 6 & 9 \end{bmatrix}$
 $B = \begin{bmatrix} 10 & 13 \\ 9 & 6 \end{bmatrix}$

$C = AB \Rightarrow C = \begin{bmatrix} 2 \times 10 + 3 \times 9 & 2 \times 13 + 3 \times 6 \\ 6 \times 10 + 9 \times 9 & 6 \times 13 + 9 \times 6 \end{bmatrix}$

$C = \begin{bmatrix} 2 \times 10 + 3 \times 9 & 2 \times 13 + 3 \times 6 \\ 6 \times 10 + 9 \times 9 & 6 \times 13 + 9 \times 6 \end{bmatrix}$



- matrix size should be $n \times n$ for both
- n should be power of 2
- $2 \times 2, 4 \times 4, 8 \times 8$



$$A = \begin{bmatrix} A_{11} & A_{12} \\ A_{21} & A_{22} \end{bmatrix} \quad B = \begin{bmatrix} B_{11} & B_{12} \\ B_{21} & B_{22} \end{bmatrix}$$

$$C = AB = \begin{bmatrix} C_{11} & C_{12} \\ C_{21} & C_{22} \end{bmatrix}$$

$C_{11} = A_{11} \times B_{11} \oplus A_{12} \times B_{21}$
 $C_{12} = A_{11} \times B_{12} \oplus A_{12} \times B_{22}$
 $C_{21} = B_{11} \times A_{21} \oplus A_{22} \times B_{21}$
 $C_{22} = A_{21} \times B_{12} \oplus A_{22} \times B_{22}$

$O(n^2)$

$\frac{n}{2}$

DAC

$T(n) = 8T\left(\frac{n}{2}\right) + 4\left(\frac{n}{2}\right)^2$

$T(n) = 8T\left(\frac{n}{2}\right) + \frac{n^2}{4}$

$T(n) = 8T\left(\frac{n}{2}\right) + n^2$

with DAE $\Rightarrow$ $\boxed{T(n) = \Theta(T(n/2)) + 4(\frac{n}{2})^2}$

$\downarrow$
 $T(n) = \Theta(T(n/2)) + 4 \frac{n^2}{4}$

$T(n) = \Theta(T(n/2)) + n^2$

$\cancel{\Theta(n^3)}$

$\swarrow$
 $f(n)$

n^2

$n^{\log_b a}$

$n^{\log_2 8}$

$\frac{n^{\sim}}{n^3}$
 $\Theta(n^3)$

according to staircase.

$$T(n) = 8T(n/2) + 4\left(\frac{n}{2}\right)^2$$

↓

$$T(n) = 7T(n/2) + 10\left(\frac{n}{2}\right)^2$$

$$T(n) = 7T(n/2) + \frac{10}{4}n^2$$

$$T(n) = 7T(n/2) + 4.5n^2$$

$$f(n) \mid n^{\log_2 7}$$

$$4.5n^2 \mid n^{\log_2 7}$$

$$n^2 \mid n^{\log_2 7}$$

$$\Theta(n^{\log_2 7})$$

$$\Theta(n^{2.81})$$

and to staircase

Ans:

$$\begin{aligned} T(n) &= T(n/2) + 4.5n^2 \\ &\searrow \\ T(n) &= O(n^{\log_2 7}) \\ &= O(n^{2.01}) \\ T(n) &= O(n^2) \end{aligned}$$

DAC

(Ogehrn)

ma df

